

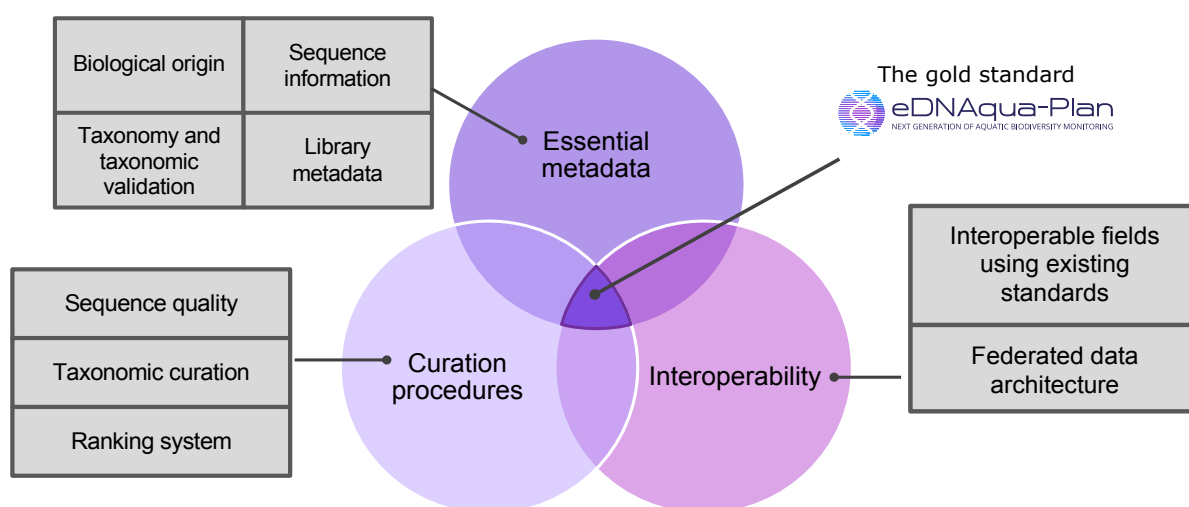
RECOMMENDATIONS FOR A FEDERATED SYSTEM OF TAXONOMICALLY CURATED REFERENCE LIBRARIES



Species identification from environmental DNA (eDNA) sequencing critically depends on the availability and quality of reference sequences to which to compare the obtained environmental sequences. For eDNA-based taxonomic identification to be routinely implemented in aquatic monitoring programs, these taxonomic assignments should be undertaken with quality-controlled and taxonomically curated reference libraries to minimise erroneous identifications. These reference libraries should additionally be easy to find and reuse. The eDNAqua-Plan deliverable report D3.1 outlines a federated network of curated reference libraries to support reproducible and reliable molecular aquatic biodiversity monitoring in Europe and beyond.

The Vision

- A federated system of taxonomically curated reference libraries
- Libraries are independently curated and maintained,
- adhere to common principles of metadata publishing, quality control, and taxonomic curation,
- and connected within a common network enabling information exchange, interoperability, and findability.
- Independence will maintain the commitment of taxonomy experts to their libraries,
- federation will bring flexibility and sustainability to the system.
- Recommendations are built around three main pillars (curation, metadata, interoperability)
- Recognition of libraries adhering to the three pillars allows for integration in the network and paves the way towards sustainable funding



Essential metadata

- **Metadata type: biological origin**
Source sample identifier, Repository, Sample type, Voucher type, Date of collection, Geographical coordinates, Country, Source sample identification, Original identification method, Type material, Project link
- **Metadata type: Sequence quality**
Sequence accession number, Source database, Source database version, Marker/genomic region, Sequencing platform, Sequencing instrument, Quality metrics, Quality control notes
- **Metadata type: Taxonomy & taxonomic validation**
Curation status, Curator/validator, Date of expert validation, Curation method, Confidence level score, Scientific name, Taxonomic identifier, Taxonomic source, Taxonomic rank
- **Metadata type: Library metadata**
Library name, Library version number, Library version date, Library marker region coverage, Library geographic coverage, Library taxonomic coverage, Library taxonomic system, Library publication, Library link, Library total records

Curation Procedures

- **Sequence quality must be verified**
Transparent, reproducible, and marker-specific criteria should be defined *a priori* and applied consistently [Ambiguous bases ("N"), Sequence length, Reading frame integrity (coding genes), Indels (coding regions), Alignment coherence, Marker confirmation (rRNA genes), Contamination screening]
- **Taxonomic names must be harmonised within a given phylogenetic clade or among identical or near-identical sequences**
One or more dedicated tools and procedures can be employed to harmonise and validate taxonomic names across the reference library:
 - Expert curation based on phylogenies
 - Expert curation based on self assignment
 - Expert curation based on clustering genetically similar sequences
 - Automated curation procedures based on bioinformatics or AI
- **Taxonomic assignments must be evaluated and assigned a quality ranking to reflect the level of confidence and reliability**
To promote harmonization across reference libraries, we recommend adopting a minimum, transparent ranking scheme grounded primarily in taxonomic traceability and morphological verification.
 - Level 1** - Sequence derived from type specimen or type strain.
 - Level 2** - Non-type specimen/strain collected near the type locality, with published morphological information
 - Level 3** - Non-type specimen/strain collected distant from the type locality, with published morphological information
 - Level 4** - Morphological data insufficient to unambiguously confirm identification
 - Level 5** - Insufficient morphological data